

Internship Report

Your Name

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1 Introduction

- Biological motivation
- Why using conformal prediction makes sense
- Outline of the work

2 Background & State of the Art

2.1 Host Prediction Problem

- problem background

2.2 State of the Art

2.3 Conformal Prediction

- framework
- coverage guarantee
- exchnagebility

2.4 The CSA Method

- ensemble score aggregation
- calibration
- envelope

2.5 Related Work???????

3 Methods

3.1 Problem Formalization

- Prediction sets - Scores - Envelope design

3.2 Envelope Contruction

3.2.1 CSA Calibration

3.2.2 Tau Computation

- one calculation, not binary search

3.3 Three envelope construction methods

- using conconvex envelopes

3.3.1 Radial Method

3.3.2 Strip Method

3.3.3 1D

3.4 $\hat{t}$ Calculation

- class conditional (why and how)

4 Data

4.1 Data Description

4.2 Protein to Phage Aggregation

- mean pooling - NaN handling

4.3 Train-Test Split

- handling the bias (currently working)

5 Experiments and Results

5.1 2D Synthetic Data

- correlated, uncorrelated etc predictors - K dimensions - multiple hosts

5.2 Gram Type Classification

using the three envelopes - Results: coverage by class, set size distribution, singleton accuracy - analysis of hard cases: ambiguous gram-positive phages

6 Conclusion and Future Work

6.1 Summary.....

6.2 Limitations

- gram-pos coverage -